📝 Mechanics Basics — Assignment Questions

A. Multiple Choice Questions (MCQs)

1. **Which of the following describes uniform motion?**
    1. Speed increases over time
    2. Speed decreases over time
    3. Speed remains constant
    4. Direction keeps changing
2. **Velocity is defined as:**
    1. Distance per unit time
    2. Speed in a given direction
    3. Change in speed
    4. Force per unit mass
3. **Acceleration is:**
    1. Change in distance over time
    2. Change in velocity over time
    3. Total displacement
    4. Rate of doing work
4. **The SI unit of force is:**
    1. Joule
    2. Watt
    3. Newton
    4. Pascal
5. **Newton's Third Law states:**

1. F = ma
2. Every action has an equal and opposite reaction
3. Objects remain at rest unless acted on

4. Energy cannot be created or destroyed

B. Short Answer Questions

1. Define **motion**.
2. State the formula for **speed**.
3. What is the difference between **speed** and **velocity**?
4. Write the formula for **acceleration**.
5. What is meant by **inertia**?

C. Calculation Questions

1. A car travels 120 km in 2 hours.
   Calculate its speed.
2. A runner changes velocity from 2 m/s to 10 m/s in 4 seconds.
   Find the acceleration.
3. A force of 15 N acts on a 3 kg object.
   Calculate the acceleration.
4. A box is pushed with a force of 20 N for 5 meters.
   Calculate the work done.
5. A 4 kg object is moving at 6 m/s.
   Find its kinetic energy.

D. Conceptual Questions

1. Explain Newton's Second Law in your own words.
2. Give one real-life example of Newton's Third Law.
3. What happens to an object if no external force acts on it?
4. Describe the difference between **kinetic** and **potential** energy.
5. Why is power considered the "rate of doing work"?

E. Extended Response

1. A student says: "If an object is moving, there must be a force pushing it."

Using Newton's First Law, explain why this statement is incorrect.

2. Describe a situation where all three of Newton's Laws apply at the same time.

F. Challenge Question

1. A 2 kg object accelerates at 3 m/s^2.
   (a) Calculate the force.
   (b) If the same force is applied to a 4 kg object, what will its acceleration be?